Objectives of Animal Breeding

Animal breeding aims at improving the genotypes of domesticated animals to increase their yield and improve the desirable qualities to produce milk, egg and meat.

When breeding takes place between animals of the same breed, it is called **inbreeding**. The cross between different breeds is called **outbreeding**.

20.4.1 Inbreeding

Inbreeding refers to the **mating of closely related animals within the same breed** for about 4-6 generations. Superior males and superior females of the same breed are identified and mated in pairs. It helps in the accumulation of superior genes and elimination of genes which are undesirable.

Hissardale is a new breed of sheep developed in Punjab by crossing Bikaneri (Magra) ewes and Australian Marino rams.

Inbreeding depression: Continued inbreeding reduces fertility and productivity. Inbreeding exposes harmful recessive genes that are eliminated by selection.

20.4.2 Outbreeding

It is the **breeding of unrelated animals**. The offsprings formed are called hybrids. The **hybrids** are stronger and vigorous than their parents. Cross between two different species with desirable features of economic value are mated. Let's see what cross produce a mule.

Cross breeding

Male Donkey X **Female Horse**

↓

Mule

Mule is superior to horse in strength, intelligence, ability to work and resistance to diseases but they are sterile.

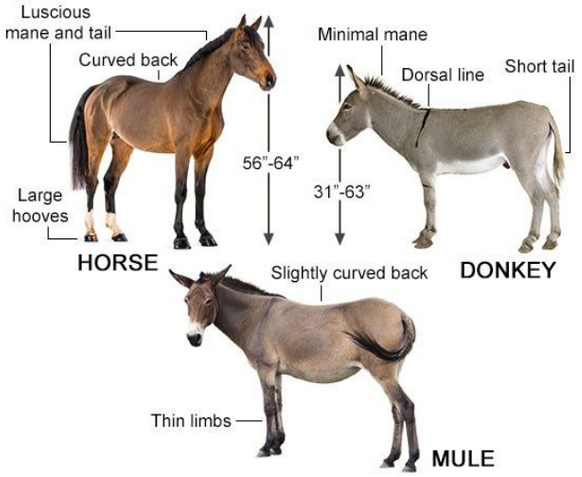


Figure 20.6 Cross breeding to produce Mule with superior characters

Info bits

Cross breed of fowls:

White Leghorn X Plymouth Rock

↓

Hybrid fowl - yield more eggs

Cross breed of cows:

Developed by mating the bulls of exotic breeds and cows of indigenous breeds.

Brown Swiss X Sahiwal

↓

Karan Swiss - yield 2-3 times more milk than indigenous cows.

20.4.3 Heterosis

The superiority of the hybrid obtained by cross breeding is called as **heterosis** or **hybrid vigour.**

Effects of hybrid vigour in animal breeding

- Increased production of milk by cattle
- Increased production of egg by poultry
- High quality of meat is produced
- Increased growth rate in domesticated animals

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